

Formulário para Cálculo II e Cálculo Aplicado II

Tabela de Derivadas

$f(x)$	$f'(x)$
c	0
x^n	nx^{n-1}
e^x	e^x
a^x	$a^x \ln(a)$
$\ln(x)$	$\frac{1}{x}$
$\log_a(x)$	$\frac{1}{x \ln(a)}$
$\text{sen}(x)$	$\cos(x)$
$\cos(x)$	$-\text{sen}(x)$
$\text{tg}(x)$	$\sec^2(x)$
$\cot(x)$	$-\csc^2(x)$
$\sec(x)$	$\sec(x) \text{tg}(x)$
$\csc(x)$	$-\csc(x) \cot(x)$
$\arctg(x)$	$\frac{1}{x^2 + 1}$
$\text{arccotg}(x)$	$\frac{-1}{x^2 + 1}$
$\arcsen(x)$	$\frac{1}{\sqrt{1-x^2}}$
$\arccos(x)$	$\frac{-1}{\sqrt{1-x^2}}$

Regras de Derivação

$$(uv)' = u'v + uv'$$

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$$

$$[f(g(x))]' = f'(g(x)) \cdot g'(x)$$

Regra da Cadeia

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

$$\frac{\partial z}{\partial u} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial u}$$

$$\frac{\partial z}{\partial v} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial v}$$

Classificação de pontos críticos

Para um ponto crítico (a, b) :

$$D = f_{xx}(a, b)f_{yy}(a, b) - (f_{xy}(a, b))^2.$$

- Se $D > 0$ e $f_{xx}(a, b) > 0$: mínimo local.
- Se $D > 0$ e $f_{xx}(a, b) < 0$: máximo local.
- Se $D < 0$: ponto de sela.
- Se $D = 0$: teste inconclusivo.

Vetor Gradiente

$$\vec{\nabla} f(x, y) = (f_x(x, y), f_y(x, y))$$

$$\vec{\nabla} f(x, y, z) = (f_x(x, y, z), f_y(x, y, z), f_z(x, y, z))$$

Plano Tangente

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

Aproximação Linear

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

$$L(x, y, z) = f(a, b, c) + f_x(a, b, c)(x - a) + f_y(a, b, c)(y - b) + f_z(a, b, c)(z - c)$$

Derivadas Direcionais

$$D_{\mathbf{u}}f(x, y) = f_x(x, y)u_1 + f_y(x, y)u_2$$

$$D_{\mathbf{u}}f(x, y, z) = f_x(x, y, z)u_1 + f_y(x, y, z)u_2 + f_z(x, y, z)u_3$$